

Information Retrieval Assignment Report-

HW8

This homework implements two fundamental unsupervised learning techniques for text analysis:

- **Topic Modeling** using Latent Dirichlet Allocation (LDA) to discover hidden thematic structure in documents
- **Document Clustering** using K-Means to group similar documents together
- **Evaluation** of clustering quality using Adjusted Rand Index (ARI) and Normalized Mutual Information (NMI)

The goal is to automatically discover topics from a large text corpus and evaluate how well unsupervised clustering matches human-labeled categories.

2. Dataset

The **20 Newsgroups** dataset was used for this assignment since we don't have AP89 dataset. This is a standard benchmark dataset for text classification and clustering.

Property	Value
Source	Kaggle (http://qwone.com/~jason/20Newsgroups/)
Total documents	37,432
Categories	20 newsgroups
Document type	Usenet messages

Documents per Newsgroup

Newsgroup	Count	Newsgroup	Count
alt.atheism	1,595	rec.sport.hockey	1,982
comp.graphics	1,933	sci.crypt	1,980
comp.os.ms-windows.misc	1,954	sci.electronics	1,952
comp.sys.ibm.pc.hardware	1,958	sci.med	1,968
comp.sys.mac.hardware	1,908	sci.space	1,964
comp.windows.x	1,944	soc.religion.christian	1,992
misc.forsale	1,898	talk.politics.guns	1,816
rec.autos	1,970	talk.politics.mideast	1,870
rec.motorcycles	1,978	talk.politics.misc	1,542
rec.sport.baseball	1,978	talk.religion.misc	1,250

3. Methodology

3.1 Text Preprocessing

Before applying LDA and clustering, the following preprocessing steps were performed:

Step	Description
Text extraction	Extracted message content from raw newsgroup files
Case normalization	Converted all text to lowercase
Stopword removal	Removed common English stopwords (using scikit-learn's list)
Vocabulary filtering	Kept words appearing in ≥ 2 documents and $\leq 95\%$ of documents
Feature selection	Limited vocabulary to 5,000 most frequent terms

3.2 Document-Term Matrix

A document-term matrix was created using CountVectorizer:

Parameter	Value	Justification
max_features	5,000	Limit dimensionality, keep most informative terms
stop_words	'english'	Remove common words with no predictive power
min_df	2	Ignore words appearing in only 1 document
max_df	0.95	Ignore words appearing in $>95\%$ of documents

Resulting matrix: 37,432 documents $\times$ 5,000 terms

3.3 Topic Modeling with LDA

Latent Dirichlet Allocation (LDA) was used to discover 20 latent topics:

Parameter	Value
n_components	20
max_iter	10
learning_method	'online'

random_state	42 (for reproducibility)
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How LDA works:

- Each document is represented as a mixture of topics
- Each topic is represented as a distribution over words
- The algorithm iteratively assigns words to topics to maximize coherence

3.4 Document Clustering with K-Means

After obtaining document-topic distributions from LDA, K-Means clustering was applied:

Parameter	Value
n_clusters	20
initialization	K-Means++
random_state	42
n_init	10

Why K-Means on LDA output?

- LDA reduces dimensionality from 5,000 terms to 20 topics
- Clustering in topic space is faster and more meaningful
- Each document is represented by its probability distribution over topics

3.5 Evaluation Metrics

Two metrics were used to evaluate clustering quality against true newsgroup labels:

Adjusted Rand Index (ARI):

- Measures similarity between cluster assignments and true labels
- Range: -1 to 1
- 1 = perfect agreement, 0 = random assignment
- Adjusted for chance

Normalized Mutual Information (NMI):

- Measures how much information clusters share with true labels
- Range: 0 to 1
- 1 = perfect agreement
- Normalized for entropy

4. Results

4.1 Discovered Topics

LDA extracted 20 topics. Below are the topics with their top 10 keywords:

Topic	Top Keywords
1	turkey, army, moon, blue, green, cover, escape, adl, battery, wings
2	people, don, think, say, does, know, just, like, make, way
3	edu, writes, article, com, cs, university, games, uiuc, news, cc
4	law, government, rights, state, israeli, states, right, arab, people, laws
5	jesus, israel, christ, lord, greek, man, russian, father, son, paul
6	ax, max, armenia, g9v, b8f, azerbaijan, a86, istanbul, holocaust, extermination
7	com, writes, article, edu, ca, like, toronto, don, db, just
8	gun, 000, police, car, weapons, arms, dod, bike, gas, door

9	just, like, don, know, time, think, good, said, going, people
10	windows, thanks, know, drive, use, edu, does, like, dos, problem
11	10, 00, 25, 1993, 12, 15, 11, 20, 17, 14
12	game, year, play, win, san, baseball, homosexual, won, homosexuality, period
13	file, edu, mail, available, information, image, program, ftp, files, software
14	stephanopoulos, mr, pl, 145, ms, 0d, 1d9, 34u, rocket, 0t
15	life, human, christians, people, evidence, does, problem, cause, drug, study
16	people, jews, armenian, turkish, war, armenians, world, jewish, guns, history
17	key, government, com, gov, nasa, public, clinton, chip, clipper, private
18	god, bible, christian, church, fbi, faith, believe, koresh, religion, children
19	space, use, new, team, high, used, data, power, bit, number
20	uk, ac, encryption, __, __, ed, virginia, ____, gm, pgp

4.2 Topic Interpretation

Theme	Topics	Key Words
Religion	5, 18	god, bible, jesus, christ, church, faith
Politics	4, 8, 16	law, government, gun, police, war, jews, armenian
Computers	3, 10, 13	windows, software, program, ftp, file, edu
Science/Space	19	space, nasa, data, power
Sports	12	game, win, baseball, hockey
General	2, 9	people, think, know, like, don

Conversation		
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4.3 Clustering Evaluation

Metric	Score
Adjusted Rand Index (ARI)	0.1138
Normalized Mutual Information (NMI)	0.2587

Interpretation:

- ARI = 0.1138 indicates moderate clustering quality (documents partially grouped by topic)
- NMI = 0.2587 indicates moderate agreement between clusters and true labels

4.4 Cluster Composition

Cluster	Size	Dominant Newsgroup	Purity
0	686	rec.sport.hockey	29.2 %
1	2,348	comp.sys.ibm.pc.hardware	25.8 %
2	1,662	rec.sport.baseball	13.4 %
3	2,779	talk.politics.misc	12.7 %
4	2,260	rec.motorcycles	24.2 %
5	1,361	comp.windows.x	40.0 %
6	1,918	comp.sys.mac.hardware	19.5 %
7	1,328	talk.politics.mideast	74.7 %
8	2,67	talk.politics.misc	12.5

	9		%
9	1,646	comp.os.ms-windows.misc	22.7%
10	2,443	sci.space	22.2%
11	591	talk.politics.mideast	28.8%
12	1,069	rec.autos	29.6%
13	2,531	comp.graphics	31.9%
14	1,928	sci.crypt	30.1%
15	1,876	rec.motorcycles	35.1%
16	2,107	sci.med	25.5%
17	1,704	talk.politics.guns	34.8%
18	1,835	soc.religion.christian	42.9%
19	1,281	rec.sport.baseball	24.0%

5. Analysis

5.1 Topic Interpretability

The LDA topics successfully captured meaningful themes:

- **Topic 5 and 18** clearly represent religious discussion (god, jesus, christ, bible, church)
- **Topic 8** captures gun control/politics discussion (gun, police, weapons)

- **Topic 10 and 13** represent computer/software discussion (windows, program, ftp, file)
- **Topic 16** captures Middle East/Politics discussion (jews, armenian, turkish, war)
- **Topic 19** represents space/science discussion (space, nasa, data, power)

Note: Some topics (e.g., Topic 11 with numbers only, Topic 14 with apparent noise) are less interpretable. This is common in unsupervised topic modeling with large vocabularies.

5.2 Clustering Quality Analysis

ARI = 0.1138 (Moderate)

The moderate ARI score is expected for several reasons:

Factor	Explanation
Unsupervised learning	Clustering used no labels; true labels were only for evaluation
Topic overlap	Computer newsgroups (comp.*) discuss similar technical content
Document length	Some newsgroup messages are very short, providing limited signal
Cross-posting	Same content sometimes posted to multiple newsgroups
Natural ambiguity	Some documents legitimately belong to multiple categories

Best Performing Cluster (Cluster 7):

- Achieved 74.7% purity for talk.politics.mideast
- Indicates Middle East politics vocabulary (Israel, Arab, peace, war) is highly distinctive
- The cluster successfully isolated this newsgroup

Challenging Clusters:

- alt.atheism documents appear as samples in clusters 0-7, indicating atheism content is distributed across many clusters rather than forming its own
- Multiple computer newsgroups overlap because they share technical vocabulary

5.3 Relationship Between Topics and Clusters

The pipeline uses a two-stage approach:

1. **LDA** reduces dimensionality from 5,000 terms to 20 topics
2. **K-Means** clusters documents based on their topic distributions

This approach works because:

- Documents with similar topic mixtures are grouped together
- The 20 LDA topics provide a meaningful latent representation
- Clustering in topic space is faster and more interpretable

5.4 Runtime Performance

Stage	Time
Document-term matrix creation	8.57 seconds
LDA training (20 topics, 10 iterations)	129.05 seconds
K-Means clustering	3.98 seconds
Total	~141.6 seconds (2.4 minutes)

The majority of time was spent on LDA training, which is expected for 37,432 documents.

6. Conclusion

Latent Dirichlet Allocation (LDA) extracted 20 topics from 37,432 documents, identifying themes including religion, politics, computers, sports, and science. K-Means clustering achieved an Adjusted Rand Index of 0.1138 and Normalized Mutual Information of 0.2587, indicating moderate agreement

with true newsgroup labels. The complete pipeline processed all documents in approximately 2.4 minutes.